DP-EAMxx DYCOMS-II RF02: cloud sedimentation × precip suppression

Plan, 6 Nov 2026. Prepared by Claude Code (AI assistant) on behalf of Naser Mahfouz, for Naser and Johannes.

Scope

EAMxx only (DP-EAMxx, P3/SHOC/SPA); EAM/MG2 serves only as the reference for the sedimentation knob. Extends the ACI × resolution study (scmlib `aci-res-study`): same case, domains, time steps and Nd targets. All on pm-gpu. Sub-km runs (625, 312, 156 m) are **held** until we have looked at these results.

Matrix (144 runs, 6 h each, 3D turbulence off)

factor	levels
dx	40, 20, 10, 5, 2.5, 1.25 km
Nd target (cm^{-3})	20, 50, 100, 200 (current SPA factors; diagnosed Nd runs 10–25 % high)
cloud sedimentation factor	0.5, 1, 2
precip suppression	on (KK2000 default), off (Nd exponent 0, prefactor compensated)

The (sed 1, suppression on) cells repeat the earlier ACI-study runs, which serves as a consistency check. Estimated cost: ∼10–20 node-hours, run packed in debug (≤8 nodes) and interactive allocations.

New knobs

- `p3::cloud_sedimentation_factor` (new, default 1.0) multiplies P3's mass- and number-weighted cloud-droplet fall speeds, the counterpart of MG2's `cld_sed`, which scales the Stokes prefactor `acn`. Droplet fall speeds used for riming are unchanged. Branch `mahf708/eamxx/p3-cloud-sed-factor` on top of E3SM PR #9017.
- **Precip suppression off**: P3 autoconversion is $1350\, q_c^{2.47} N_d^{-1.79}$ (N_d in cm^{-3}). We set the exponent to 0 and the prefactor to $1350 \cdot N_{\mathrm{ref}}^{-1.79}$ with $N_{\mathrm{ref}} = 63\,cm^{-3}$ (the geometric mean of 20–200), giving **0.812**. Other reference values: 1.04 at $55\,cm^{-3}$ (the default SPA Nd at DYCOMS), 1.23 at 50, 0.355 at 100. Nd still acts through sedimentation, evaporation and LW (SW is off in this nocturnal case).

New output

- **AeroCom cloud-top fields** (PR #9017 `aerocom(..)` operator), every 15 min, per column: cloud-top fraction (total and liquid), T, p, r_{eff} and in-cloud CDNC at liquid cloud top.
- **LWP spatial variance** online: `((LWP - LWP.mean('col'))**2).mean('col')`, written with `broadcast_like`. The SD is its square root, cross-checked against the 15-min 2D LWP output.

Steps

1. One build (SICE compset, resolution-independent executable); 40 km smoke test of the new knob and output.
2. Run 40 km–2.5 km, then 1.25 km; check Nd, stability and the knob's effect on rain and w_e.
3. Analysis as in the ACI study (h4–6 means, susceptibility $d\ln \mathrm{LWP}/d\ln N_d$, w_e, rain, cloud cover, LWP SD, cloud-top CDNC/r_{eff}); results emailed to both of us.
4. Sub-km only after we discuss.